Grippy: Operational Strategy Report

CVO Final Project - Operations Management (2026F-A02)

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1. Venture Overview & Value Proposition

1.1 Industry Opportunity

Electricians rarely work from one fixed site. They move between job sites with a large tool kit, often work in confined spaces and must carry, organize and reach that kit throughout the working day, which costs them time and physical effort.

Conventional upright backpacks address portability and storage, but they force electricians to dig through layered compartments for tools buried at the bottom. The closest product, Klein Tools' Tool Station Backpack, adds a front flap that folds down into a workstation and padded, adjustable straps[1]. Even there, comfort comes from padding added to a storage bag rather than from a design built around carrying. Back support has been a recurring pain point in our interviews with electricians[2]. Incumbents therefore under-serve independent electricians who carry a full kit between job sites every day: they need tool visibility and on-the-move access combined with ergonomic comfort, and no bag we reviewed offers all three.

1.2 Proposed Customer and Value Proposition

Grippy targets independent electrical contractors who work across commercial and residential job sites without a fixed workplace. The Grippy backpack leads with an ergonomic support harness designed to reduce strain when carrying a full kit between sites. It opens flat into a workspace, and its detachable compartments (a base set plus add-ons the customer picks when ordering) can be fixed onto the electrician's own belt for access on the move. The flat opening is meant to keep tools in view and cut the time lost to rummaging.

1.3 Uncommon Belief and Value Theory

The common belief in the trade is that a tool bag must trade off capacity, access and carrying comfort. Grippy's uncommon belief is that the trade-off comes from building bags around storage: a bag designed around carrying can combine capacity, access and comfort, and independent electricians will pay a premium for it. Following the Value Lab approach[3], the problem Grippy must solve is to deliver that comfort and access without giving up the capacity/durability incumbents already offer. Section 6.3 tests Grippy's value theory: independent electricians who move between job sites will pay $145, about $35 more than the Klein Tool

[1]Klein Tools, Tradesman Pro Tool Station Backpack (Cat. No. 55482), kleintools.com.
[2]Grippy team interviews with electricians, fall 2026.
[3]Felin, Gambardella and Zenger (2021), “Value Lab.”

Station, for an open-flat backpack whose detachable compartments and ergonomic support harness give on-the-move access and carrying comfort. If the theory holds, carrying comfort becomes an order winner in a category that competes on storage: independent electricians would switch first, other mobile trades could follow, and incumbents would have to rethink bags designed as storage products.

1.4 Scope and Strategic Choice

The venture will initially focus on one standardized model (one body in one size, with a base set of detachable compartments) to minimize production overhead and manufacturing complexity. A contract manufacturer will sew the bags and Grippy will do the final kitting (Section 4.1). Because Grippy expects customers to pay for design rather than a low price, it competes on Product Leadership (Section 2.2).

2. Competitive & Operational Strategy

2.1 Competitive Landscape and Positioning

Stanley Black & Decker (including DEWALT) and Klein Tools dominate professional tool storage, and their brands, distribution and reliable manufacturing set a high barrier to entry.

Stanley Black & Decker, the market leader in tools, reported $13.3 billion of Tools & Outdoor sales in fiscal 2024[4]. Two blocks of its Business Model Canvas[5] (Exhibit 1) explain its position. Its Channels (home centers, mass merchants and distributors) put its brands in front of every trade, and its Key Resources are global manufacturing scale and brands such as DEWALT. Klein Tools, a family-owned manufacturer founded in 1857, is the electrician specialist[6]. Its position rests on Customer Segments and Value Proposition, because it sells trade-specific durability to electricians. Their cost structures differ too: Stanley Black & Decker spreads scale-driven costs across every trade, while Klein accepts higher unit costs to serve electricians. Each has a vulnerability: Stanley Black & Decker's scale model favors broad, standardized lines over niche designs, and Klein's bags extend a storage product rather than rethinking how electricians carry and reach their tools.

[4]Stanley Black & Decker, Form 10-K, fiscal 2024.
[5]Osterwalder and Pigneur (2010), Business Model Generation.
[6]Klein Tools, company website, kleintools.com.

The closest direct competitor is Klein Tools' Tool Station Backpack (Section 1.1). At about $110[7], it sets the price reference for Grippy's $145. Grippy must earn that $35 premium with two features the Klein listing does not describe: the ergonomic support harness and detachable compartments for access on the move.

2.2 Value Discipline and Operational Alignment

Grippy adopts Product Leadership, one of Treacy and Wiersema's three value disciplines[8]. Their rule is to lead on one discipline and meet the industry standard on the other two. Stanley Black & Decker leads on Operational Excellence, using scale to keep costs low, while Klein builds its range and reputation around one trade. A startup making small outsourced batches cannot match the first on cost or the second on trust, but it can lead on design and charge a premium for it. In Hill's terms[9], functional design and ergonomics are the order winners that make an electrician choose Grippy. Durability, a $145 price electricians accept and a reasonable delivery time are qualifiers: Grippy must meet them to be considered but cannot win on them. Used as a decision rule, the discipline settles the operational choices that follow: iterative development (Section 3), a responsive Assemble-to-Order supply chain with outsourced sewing (Section 4) and quality defined as functional performance (Section 5).

3. New Product Development Strategy

Product Leadership puts development at the center of Grippy's operations, because the design is what customers pay for. The first version (V1) has three features: a backpack that opens flat, detachable compartments and an ergonomic support harness; Grippy's own utility belt and a hanging point wait for V2.

Our initial customer research found two linked problems: electricians struggle to reach their tools quickly while moving around a job site, and the physical strain of carrying a full kit grows as electricians get older. These findings led us to challenge the common belief (Section 1.3) that a tool bag must trade off capacity, access and carrying comfort.

The next step was a rough cardboard prototype, which opens fully flat but does not yet have the detachable compartments. Cardboard is enough to test tool retrieval time, tool visibility and how

[7]Keep Supply, Klein 55482 listing, accessed October 27, 2026.
[8]Treacy and Wiersema (1993), Harvard Business Review.
[9]Hill (1985), Manufacturing Strategy.

easily the bag opens flat, but load, durability and carrying comfort need a sewn prototype. Sections 6.2 and 6.3 set out the test protocols, which compare Grippy with a conventional backpack and the Klein Tool Station.

The capability audit in Section 6.1 shows that commercialization needs capabilities Grippy lacks in ergonomic design, manufacturing and quality assurance. Consistent with Product Leadership, development is iterative, and each prototype round targets one or more of these gaps. Leading on design does not excuse weak production: under Treacy and Wiersema's threshold rule, the bag must still be made as consistently as an incumbent's.

The prototypes are Grippy's proof of concept: they show whether the design works. A proof of production must follow before a full launch, so Grippy will end its pre-launch stage (Phase 0) with a 50-unit pilot batch that tests whether the manufacturer can make the bag repeatedly within the target defect rate and lead time. It will also expose bottlenecks and test quality control and distribution without over-committing resources. Grippy launches only if the pilot meets the triggers in Section 6.1.

4. Operational Process & Supply Chain Strategy

4.1 Proposed Operating Model

The operating model must deliver the order winners and qualifiers in Section 2.2 while the design is still changing between prototype rounds. Fisher's framework sets the type of supply chain[10]. Grippy is an innovative product: demand is unproven, the target gross margin is at least 50% and customers choose their own configuration. Innovative products need a market-responsive supply chain, which values speed and flexibility over the lowest unit cost, whereas Stanley Black & Decker's functional products suit a physically efficient one. Grippy will therefore start with small batches, component buffers and a manufacturer chosen for flexibility rather than lowest cost.

Grippy will source fabric, webbing and hardware such as zippers, buckles and connectors from suppliers. A contract backpack manufacturer will cut and sew the bag bodies, straps and compartments to Grippy's specification. Grippy will inspect incoming lots, hold component stock and do the final kitting of each order.

[10]Fisher (1997), Harvard Business Review.

4.2 Process Design

Proposed Production Workflow: Grippy sources materials → Manufacturer sews bodies, straps and compartments → Inspection 1 (manufacturer) → Inspection 2 (Grippy goods-in) → Component stock → Customer order → Grippy final kitting [CODP] → Inspection 3 (opens flat, compartments attach, harness and strap load) → Ship

Potential Inefficiencies. The likeliest are material shortages, supplier delays and poor stitching on a specialized design that takes longer than usual to sew.

4.3 Push vs. Pull Strategy

A push strategy produces ahead of demand to cut waiting times, but for a new product it risks unsold inventory. A pull strategy produces only to order, which removes that risk but lengthens waiting times.

At launch (Phase 1), Grippy combines the two through Assemble-to-Order: bodies and compartments are made ahead of demand (push), and final kitting starts only when an order arrives (pull). The boundary between push and pull is the CODP (Section 4.4).

Once demand has been stable for three months (Phase 2), Grippy also holds a small stock of the most popular configurations, which moves the CODP to finished goods for those configurations only.

4.4 Customer Order Decoupling Point (CODP)

The Customer Order Decoupling Point (CODP) is the stage where production shifts from forecast-driven to customer-order-driven, and where a firm holds its strategic inventory[11].

Three potential strategies for Grippy are Make-to-Stock (MTS), Assemble-to-Order (ATO) and Make-to-Order (MTO), compared in Exhibit 2.

Recommended Strategy: Assemble-to-Order (ATO). ATO suits Grippy's modular design, because a base set of detachable compartments plus the add-ons each customer picks when ordering creates many configurations.

Stocking every combination as a finished bag would be impractical, while Make-to-Order would lengthen delivery time, a qualifier in Section 2.2.

[11]Olhager (2003), International Journal of Production Economics.

The CODP therefore sits at final kitting, where stocked modules are built into the bag the customer ordered. Holding stock as components also postpones differentiation: Grippy forecasts one body and a few compartment types, not every configuration.

4.5 Operational Coordination

Production Coordination. Grippy will agree production schedules and standard specifications with the manufacturer, so that parts arrive when needed and quality stays consistent between batches.

Inventory Coordination. Grippy reorders bodies and compartments when stock falls to the expected demand over the replenishment lead time (materials plus sewing) plus a safety stock, so kitting does not wait for parts and inventory stays low.

Order and Fulfillment Coordination. Grippy will sell direct online to independent contractors first and through electrical supply houses later. One order management system links orders to component stock, so each confirmed order can be reserved, kitted and shipped within the target of three business days.

Outsourced sewing, in-house final kitting and Assemble-to-Order keep fixed costs and inventory risk low while letting each electrician choose a configuration.

5. Capacity, Efficiency & Quality Management

5.1 Production Capacity and Resource Planning

A sellable bag needs heavy-duty nylon or polyester that can withstand job sites, breathable mesh straps and heavy-duty nylon stitching, the most critical of the three because a bag that opens fully puts repeated stress on its seams.

Grippy will select a contract backpack manufacturer that can sew the open-flat construction and the support harness, accepts a low minimum order, quotes short lead times and has a record of consistent quality. Capacity then depends on two stages. Upstream, the manufacturer's minimum order and lead time set how many bodies and compartments Grippy must order and hold. Downstream, final kitting sets weekly output: staffed kitting minutes divided by the minutes each order takes. Until quotes arrive, Grippy plans on a 50-unit pilot, a lead time of up to eight weeks and about ten minutes of kitting per order, so ten staffed hours a week would kit about 60 orders.

The Theory of Constraints holds that the constraint sets the output of the whole system[12]. At launch the likeliest constraint is demand, not any process. Within operations, component supply binds before kitting: Grippy can only kit bodies it holds, and each replenishment takes up to eight weeks. Grippy therefore protects supply first, through the reorder points in Section 4.5, and adds kitting hours only when orders approach 60 a week.

5.2 Operational Efficiency and Capacity Improvements

Grippy's main efficiency lever is standardization. Before the pilot, it will fix the product's measurements, document its material and stitching specifications and set one attachment method for all compartments. These specifications and a signed-off sample close the interpretation gap between Grippy's design and what the manufacturer sews. They also cut production errors and keep kitting repeatable and within ten minutes per order.

Grippy will track six KPIs, three for the product and three for the process. The product targets are fixed before testing (Section 6.2): tool retrieval at least 30% faster than from a conventional backpack, no failure below 22.5 kg (1.5 times the 15 kg rated load) and none within 1,000 open/close cycles. From the pilot batch on, the process targets are a defect rate of 3% or less, final kitting within ten minutes per order and shipping within three business days. Grippy records the product KPIs in every test round and all six in the pilot.

5.3 Product Quality & Performance Measurement

Grippy's quality standard follows from its value proposition: each bag must carry its rated load, survive repeated open and close cycles and keep the harness securely attached under load. The physiotherapist check in Section 6.1 judges how well the harness spreads that load. The pilot batch then tests process capability: Grippy measures stitch density and seam dimensions on all 50 units against its specification limits, targeting a Cpk of at least 1.33, and tests seam strength destructively on five.

Feedback from electricians in the target market will show where the bag falls short of that standard and guide changes to the harness.

Grippy will inspect at the three points in the Section 4.2 workflow: the manufacturer checks each lot before shipping, Grippy checks it on arrival and every kitted order gets a final check. Recording each inspection's findings will expose recurring supplier or process issues and guide

[12]Goldratt and Cox (1984), The Goal.

design or manufacturing changes. In cost-of-quality terms, the specifications and signed-off sample are prevention costs, the inspections are appraisal costs and lots they reject are internal-failure costs. All three cost less than an external failure: a seam that splits on a job site means a return, a replacement and a lost customer for a brand that charges a premium.

6. Operational Risks & Implementation Strategy

6.1 Capability Gaps and Implementation Path

The capability audit in Exhibit 3 compares what Product Leadership requires (Section 2.2) with what Grippy has today: a cardboard prototype, interview findings and no production, testing or sales capability. Design and ergonomics must beat the Klein Tool Station by enough to justify the $35 premium, while durability only needs to match it and delivery time only needs to be reasonable. Three gaps stand between the prototype and a sellable product:

- **Ergonomic design (build, with an adviser).** The harness is the main order winner, yet no team member is trained in ergonomics. Grippy keeps design in-house, as the source of differentiation, and has a physiotherapist check how the harness spreads a full tool load.
- **Manufacturing (partner).** Sewing an open-flat body and a load-bearing harness needs industrial machines that would sit idle at Grippy's volumes. A contract manufacturer turns that fixed cost into a unit cost, at the price of less control over quality and lead time.
- **Quality assurance (build, then buy).** Durability is a qualifier: a bag that fails early loses the customer whatever its design. Grippy builds simple load and open/close tests in-house and buys certified lab tests before launch.

Price, final kitting and online sales are smaller gaps to close before launch. Phase 0 (at least three months) produces sewn samples, runs the tests in Sections 6.2 and 6.3 and ends with a 50-unit pilot batch. Grippy moves to Phase 1, Assemble-to-Order sold direct online, only if the pilot's defect rate is 3% or lower (no more than one defective bag in 50) and the manufacturer delivers it within eight weeks. After three months of stable demand, Phase 2 stocks the most popular configurations and adds electrical supply houses. Phase 0's length, the batch size and the eight-week lead time are planning assumptions until manufacturer quotes replace them.

6.2 Deal-Killer Risks and Experiments

A deal-killer is an assumption that, if false, stops the plan whatever else goes right. Grippy has four, each with a kill threshold fixed before testing (Exhibit 4). Three of them test requirements from the capability audit in Exhibit 3 (functional design, unit cost and durability); the fourth, whether electricians will pay $145, is tested in Section 6.3. The three are:

- **The access benefit is not real.** If opening flat does not make tools faster to reach, the value proposition fails. Test: the rapid prototype test below. Kill threshold: Grippy is less than 30% faster than a conventional backpack.
- **Unit cost is too high.** Product Leadership lets Grippy plan a $145 price, but that is already about a third above Klein's $110, so higher costs cannot be passed on. Test: quotes from three contract manufacturers, plus material and freight costs. Kill threshold: a landed unit cost above $72.50 at 500 units. That would leave a gross margin below 50%, too thin to fund marketing, shipping and returns.
- **The bag cannot carry a full kit.** Opening flat puts a long seam and zipper under load. Test: load and open/close cycle tests on three sewn samples. Kill threshold: any failure below 1.5 times the 15 kg rated load or within 1,000 cycles (about a year at three openings a day).

Rapid prototype test. Five participants each retrieve the same five tools from a conventional backpack and from the cardboard Grippy opened flat, alternating which bag comes first. Each retrieval is timed and dropped tools are counted. The baseline is a conventional backpack because Section 1.1 makes the access claim against upright bags; Section 6.3 compares Grippy with Klein.

Path-dependent risks. These differ from deal-killers: resolving them changes the venture's direction rather than ending it. Early orders, for example, will show which add-on compartments sell, which decides the configurations Phase 2 stocks and where the CODP moves. Grippy tests the deal-killers first: they are cheap to test before the pilot and fatal if found after it.

6.3 Value-Proposition Test

The value proposition is the ultimate deal-killer: if electricians will not pay, no operational plan can save the venture. Hypothesis: the value theory in Section 1.3, that independent electricians will pay $145 for Grippy's carrying comfort and on-the-move access. Method: 15 independent

electricians try a loaded sewn sample and a loaded Klein Tool Station, with the Klein's $110 price shown. Grippy asks which they would buy at $145 and the most they would pay. It then offers a refundable $20 pre-order deposit to separate polite interest from commitment. Kill threshold, set in advance: fewer than half prefer Grippy at $145 or fewer than three pay a deposit.

7. Strategic Recommendations & Conclusion

Grippy should compete on Product Leadership and shape its operations around that choice. First, launch one $145 V1 backpack that opens flat, with detachable compartments and an ergonomic support harness. Hold Grippy's own utility belt and the hanging point for V2 so that development stays on the harness, the main order winner. Second, outsource sewing but keep design and final kitting in-house, because that is where differentiation and the customer order decoupling point sit. Third, run a responsive Assemble-to-Order supply chain from launch and sell direct online for margin and customer feedback, changing phase only when the triggers in Section 6.1 are met.

The tests form a go/no-go sequence that moves from proof of concept to proof of production. Grippy orders the pilot batch only if all four tests clear their kill thresholds, and launches only if the pilot meets the triggers in Section 6.1. Because each test costs little next to a production order, Grippy runs the tests first and makes its largest commitment last.

Today's tool bags force electricians to trade comfort and access for capacity. Grippy bets they will pay to avoid that trade-off. It competes on Product Leadership, builds a modular Assemble-to-Order operation around its design and tests each deal-killer before ordering the pilot batch.

Exhibits

Exhibit 1. Business Model Canvas: Stanley Black & Decker and Klein Tools

Block	Stanley Black & Decker (incl. DEWALT)	Klein Tools
Customer Segments	All trades and DIY consumers	Professional electricians and allied trades
Value Proposition	Broad, reliable tool and storage range at competitive prices	Trade-specific durability and electrician-focused design
Channels	Home centers, mass merchants, distributors, online retail	Electrical distributors, home centers, online retail
Customer Relationships	Brand loyalty built through tool platforms	Long-standing trust among electricians
Revenue Streams	Power tools, hand tools, storage and outdoor products	Hand tools, testers, tool bags and storage
Key Resources	Global manufacturing scale; brands such as DEWALT, STANLEY and CRAFTSMAN	Brand reputation with electricians; specialist product design
Key Activities	High-volume manufacturing, supply-chain management, marketing	Product design and quality manufacturing for the electrical trade
Key Partners	Large retailers, suppliers, contract manufacturers	Electrical distributors and suppliers
Cost Structure	Scale-driven, low unit cost	Higher unit cost, premium positioning

Decisive blocks: Channels and Key Resources for Stanley Black & Decker; Customer Segments and Value Proposition for Klein Tools. Template and sources: footnotes 4 to 6.

Exhibit 2. Customer order decoupling point options for Grippy

Strategy	Application to Grippy	Advantages	Limitations
Make-to-Stock (MTS)	Finished Grippy backpacks are manufactured and stored before orders arrive.	Fast delivery and efficient batch production.	Higher inventory risk and limited flexibility.
Assemble-to-Order (ATO)	Standard bag components are made in advance and kitted after an order is received.	Balances customization, inventory control and delivery speed.	Requires component availability and efficient assembly coordination.
Make-to-Order (MTO)	Manufacturing begins only after a customer places an order.	Low finished-goods inventory and greater customization.	Longer delivery times and potentially higher unit costs.

Exhibit 3. Strategic operational audit and capability gaps

Requirement (Section 2.2)	Current capability	Gap	Decision and reason
Ergonomic harness that beats Klein (order winner)	Interview findings only	Large	Build, with a physiotherapist adviser: the harness is the differentiator
Open-flat body and detachable compartments sewn to specification (order winner)	None	Large	Partner with a contract manufacturer: machines too costly at low volume
Durability equal to Klein (qualifier)	No standards or test equipment	Large	Build simple tests; buy certified lab tests before launch
$145 price electricians accept (qualifier)	No cost or willingness-to-pay data	Medium	Collect three manufacturer quotes; test $145 with 15 electricians
Reasonable delivery time (qualifier)	No kitting or fulfillment process	Small	Build in-house: kitting is simple and holds the CODP
Direct online sales (Section 4.5)	None	Small	Buy a standard e-commerce platform

Exhibit 4. Experiment protocols and kill thresholds

Deal-killer	Test	Prototype	Sample	Kill threshold
Access benefit not real	Timed tool retrieval against a conventional backpack	Cardboard (now)	5 people, 5 tools each	Grippy less than 30% faster
Unit cost too high	Manufacturer quotes	Technical drawings	3 manufacturers	Landed cost above $72.50 at 500 units
Bag cannot carry a full kit	Load and open/close cycle tests	Sewn samples	3 samples	Failure below 22.5 kg (1.5 × 15 kg) or within 1,000 cycles
Value proposition fails	Side-by-side with Klein; refundable pre-order deposit	Sewn sample and a Klein Tool Station	15 electricians	Under half prefer Grippy at $145 or fewer than 3 deposits

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